

AI for Bharat Hackathon

Powered by **aws**



Team Name : GenGurus

Team Leader Name : Vedika Mishra

Problem Statement : **[Professional Track] AI for Retail, Commerce & Market Intelligence**

Brief about the Idea:

The problem

India's millions of MSMEs—including kirana stores and small retailers—often run on intuition and experience. They face stock-outs during peak demand, miss sales opportunities, and lack data-driven guidance, leading to lost revenue and inefficient inventory.

The idea-

AI Sahayak is a proactive intelligence platform built for MSMEs and small retailers. Instead of the owner searching for answers, the system uses AWS EventBridge (Change Calendar), Amazon SageMaker, and Amazon Bedrock to predict demand, detect anomalies, and send actionable alerts (e.g. via WhatsApp) at the right time—in Hinglish, so it feels natural.

In one line

“AI that reaches out to the MSME owner before they have to ask—so they can focus on the business, not the spreadsheet.”



Why AI is required in your solution?

Proactive intelligence: MSMEs can't manually track calendar events, sales patterns, and inventory. AI predicts demand and risk so we can *push* the right insight at the right time instead of the owner searching for it.

Demand forecasting: Historical data + events (festivals, holidays) need ML (e.g. SageMaker) to estimate what will sell, when—so owners can stock and order smarter.

Natural, inclusive interaction: Owners prefer speaking in Hinglish. AI (e.g. LLMs on Bedrock) powers conversational alerts and chat so the experience is natural and accessible.

Anomaly detection: AI spots unusual patterns (e.g. sudden drop in sales, stock-outs) and triggers alerts so owners can act before the problem grows.

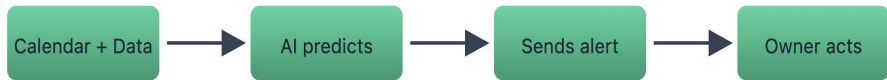
One line:
AI is required to turn raw data and events into proactive, timely, and understandable guidance for MSME owners.

Without AI (Reactive)



Owner has to remember to check and search

With AI Sahayak (Proactive)



How AWS services are used within your architecture?

Amazon EventBridge + Change Calendar: Drives proactive behaviour. Calendar events (festivals, holidays) trigger pipelines so we generate and send alerts at the right time.

Amazon SageMaker: Demand forecasting and/or anomaly detection models; training and deployment so predictions are scalable and accurate.

Amazon Bedrock: LLMs for Hinglish alerts, chat, and summarisation so messages are natural and easy to act on.

AWS Lambda: Serverless orchestration—e.g. reacting to EventBridge, calling SageMaker/Bedrock, and integrating with WhatsApp or other channels.

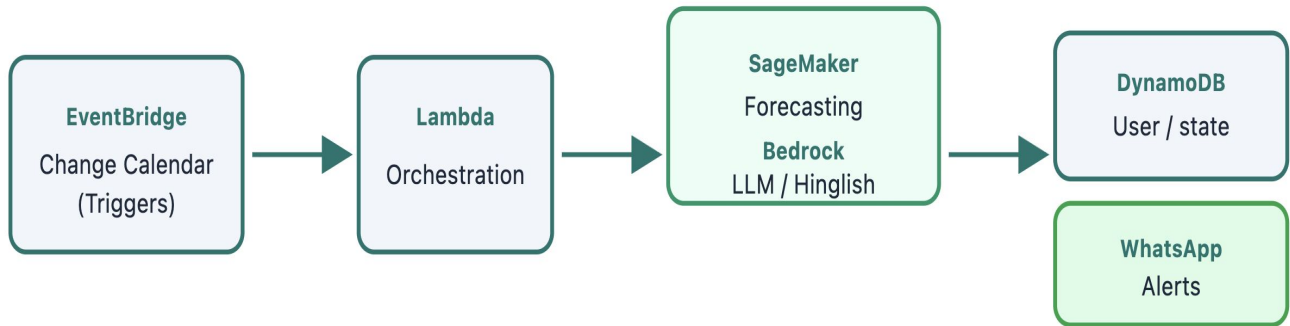
Amazon DynamoDB: User profiles, preferences, alert settings, and session/chat state for personalised, persistent experience.

Amazon Cognito: Secure sign-in for the web app; identity for API and personalisation.

Amazon S3: Storage for config, panchang/calendar data, or other static assets used by the AI pipeline.

One line:

EventBridge and Change Calendar make the system proactive; SageMaker and Bedrock add intelligence; Lambda, DynamoDB, Cognito, and S3 make it secure, scalable, and personalised.



What value the AI layer adds to the user experience?

“Alerts find the user”: Instead of opening an app and searching, the owner gets timely, relevant messages (e.g. WhatsApp) in Hinglish—so they can act without learning a complex tool.

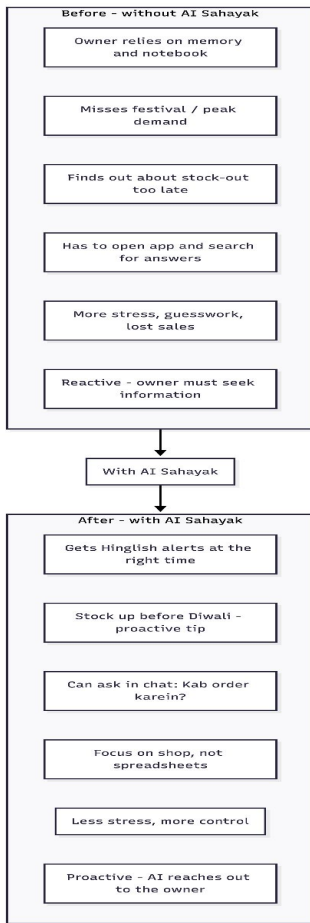
Conversational support: In the app or chat, they can ask in their own words (e.g. “Kab stock order karna chahiye?”). The AI layer understands intent and returns clear, actionable answers.

Relevant to their context: AI uses their location, calendar (festivals, holidays), and history so suggestions and alerts feel specific to their shop and time.

Less stress, more control: Predictions and anomaly alerts reduce guesswork and surprise stock-outs, so the owner feels more in control and less worried.

One line:

The AI layer turns data and events into a simple, conversational, and proactive experience—so MSME owners spend time on the business, not on spreadsheets or complex dashboards.



List of features offered by the solution

Proactive alerts — Timely, event-driven alerts (e.g. festival reminders, stock-up suggestions) via WhatsApp or in-app, so the owner doesn't have to search.

Demand forecasting — ML-based demand predictions (e.g. SageMaker) so owners can plan inventory and orders.

Hinglish conversational AI — Chat and alerts in Hinglish (e.g. Bedrock) so MSME owners can ask in their own words and get clear, actionable answers.

What-If simulation — Slider or scenario tool to explore “what if I change price or stock?” for quick decision support.

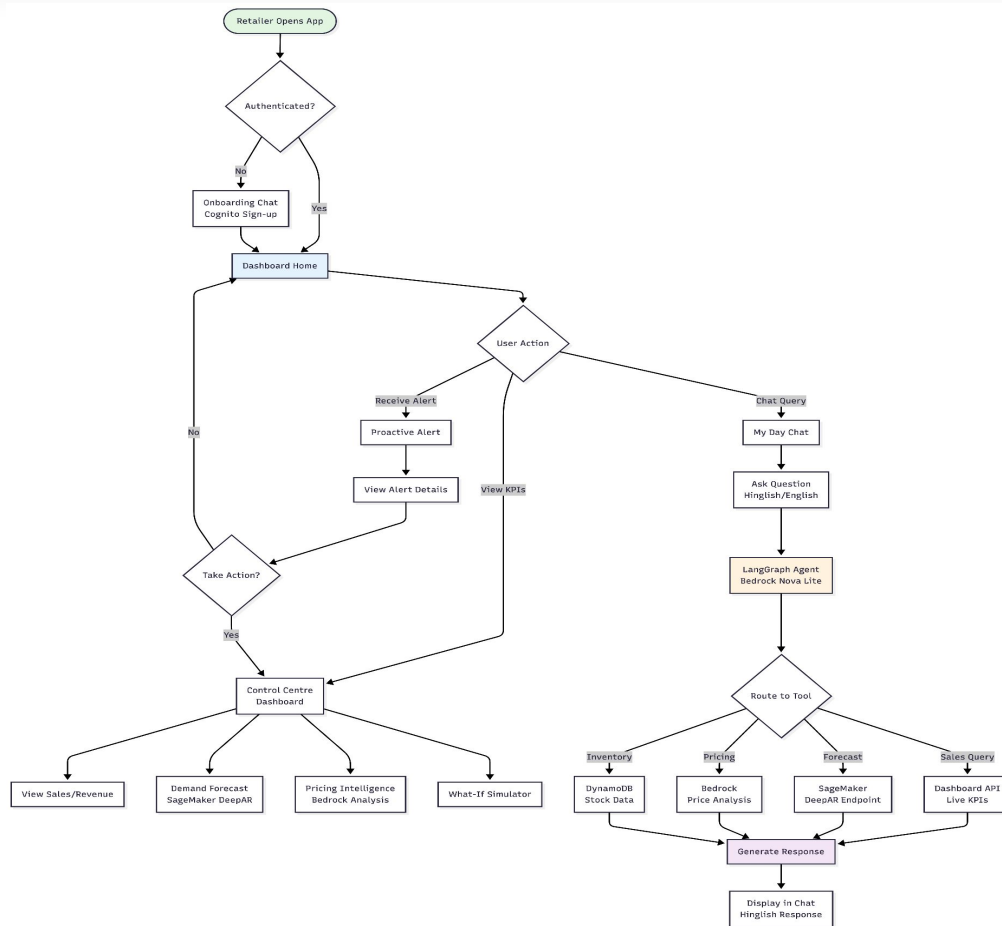
My Day / Live Alerts — Daily summary and live alerts in one place; optional alert-time preference (e.g. “set alert for 4 pm”) stored in DynamoDB.

Panchang / calendar-aware logic — Regional festivals and calendar events (e.g. S3 panchang data) used to trigger relevant alerts and suggestions.

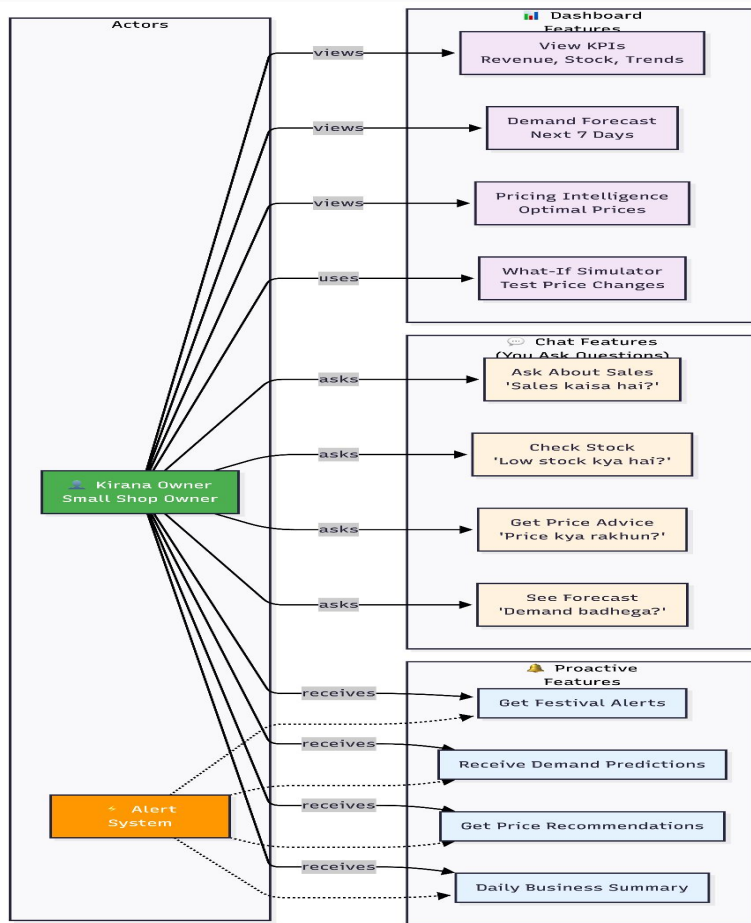
Personalised dashboard — Overview of sales, trends, and recommendations tied to the signed-in user (Cognito + DynamoDB).

Secure, scalable backend — Auth via Cognito; serverless orchestration (Lambda); data in DynamoDB and S3 so the solution scales for many MSMEs.

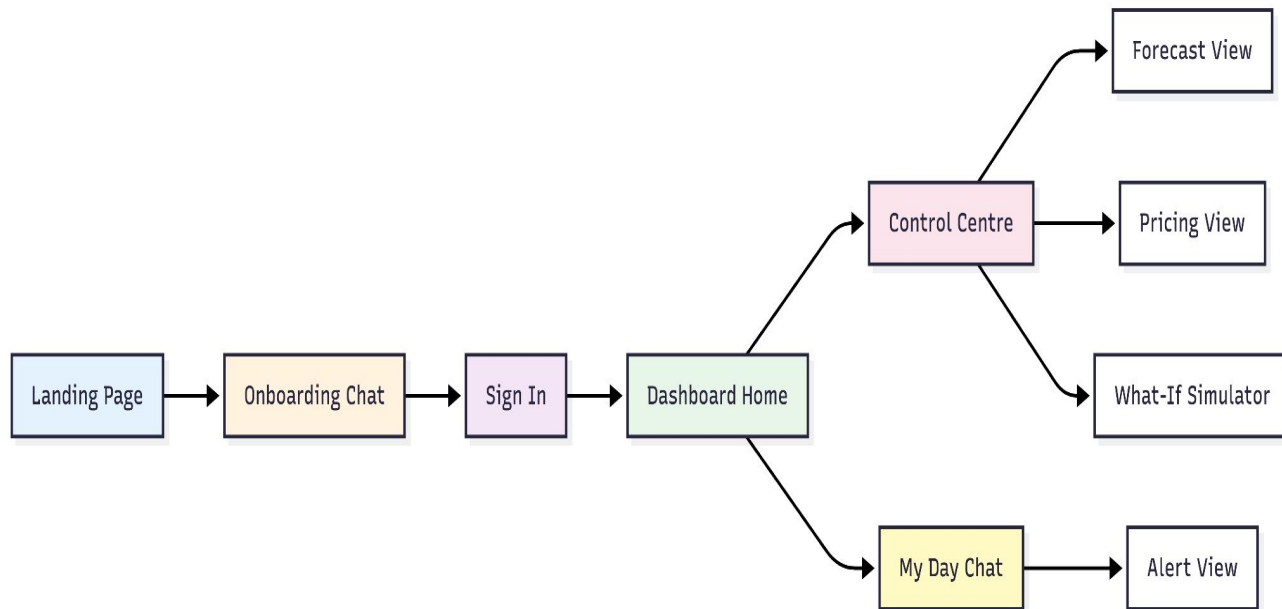
Process flow diagram



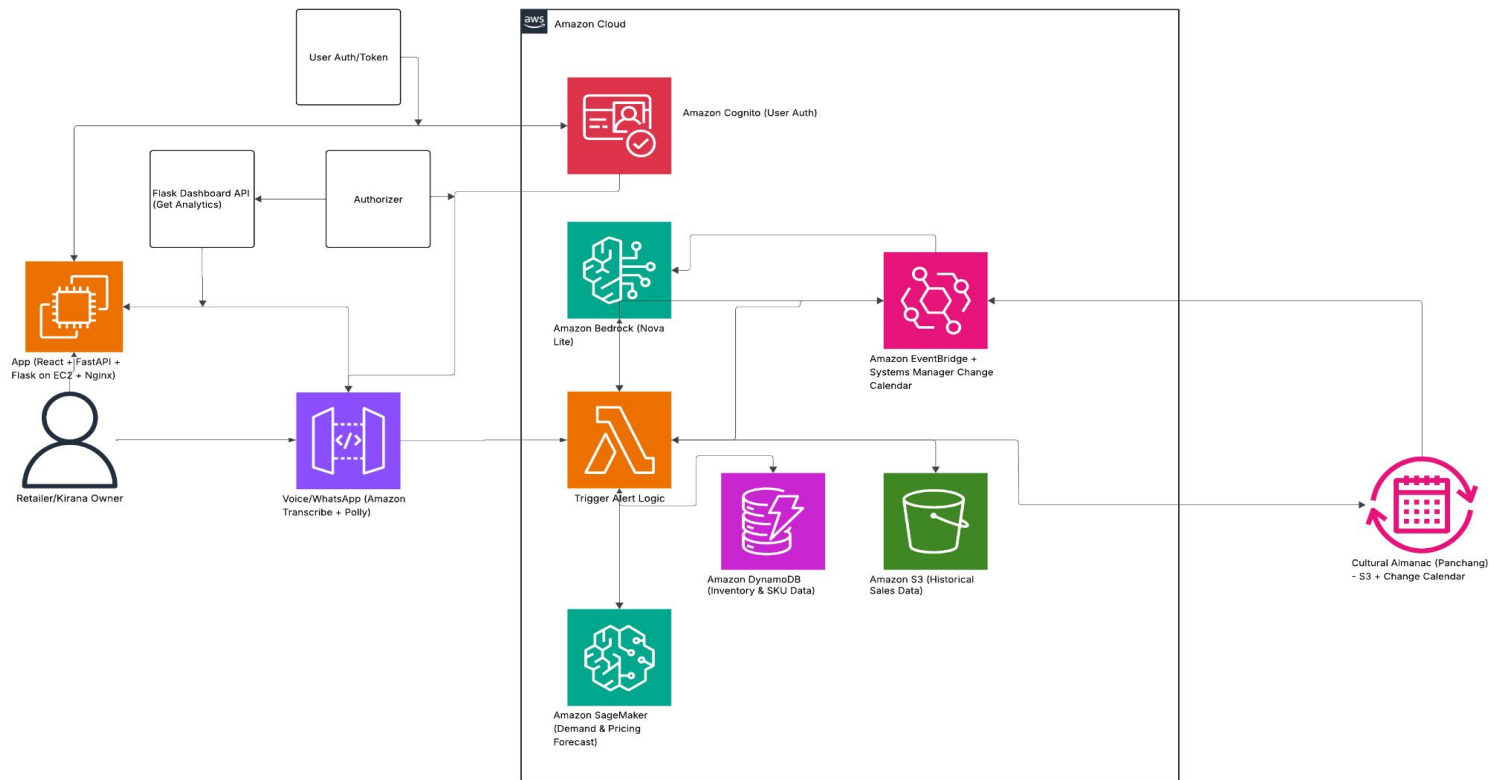
Use-case diagram



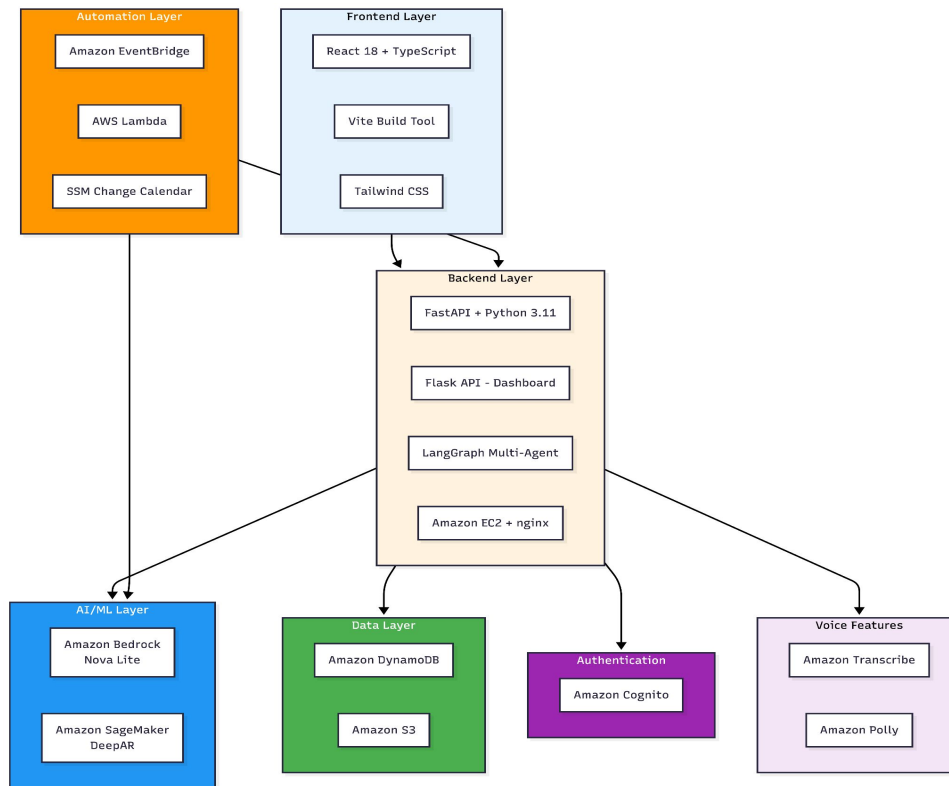
Wireframes/Mock diagrams of the proposed solution



Architecture diagram of the proposed solution:



Technologies utilized in the solution:



Estimated implementation cost:

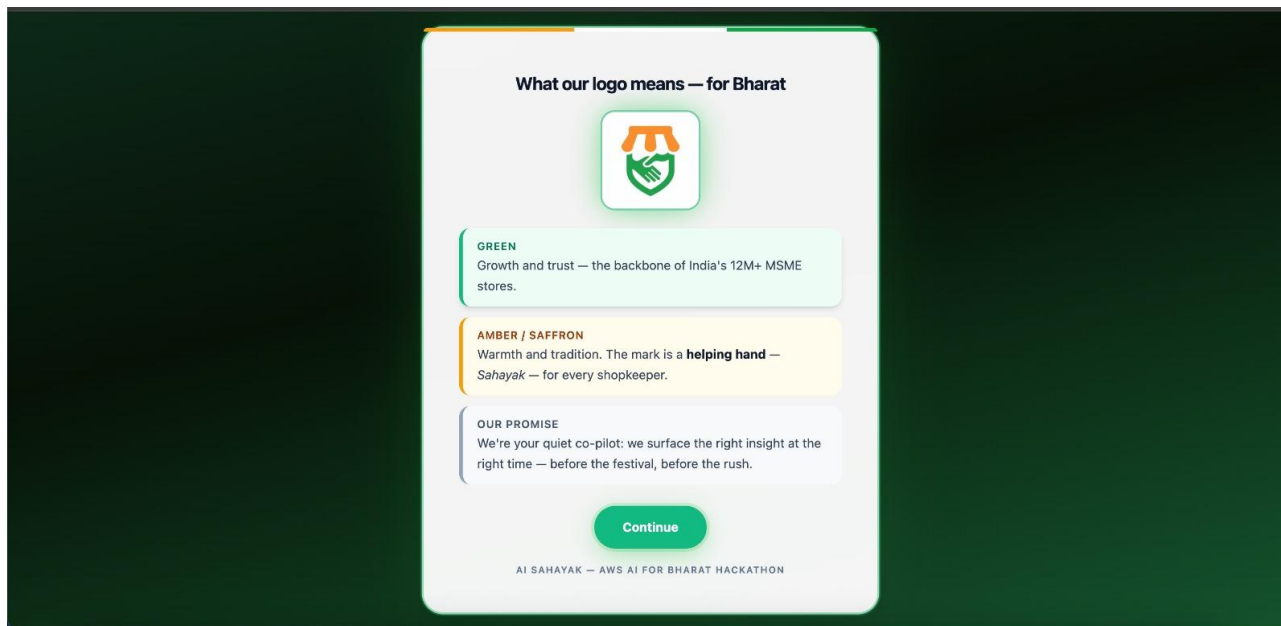
[AI-Sahayak AWS Cost](#)

Estimate summary			
Upfront cost	Monthly cost	Total 12 months cost	Currency
205.2	36.58	644.16	USD
		* Includes upfront cost	

Service	Upfront	Monthly	First 12 months total	Currency
Amazon	0	0.0051	0.06	USD
Anthropic	0	0.0864	1.04	USD
Amazon Cognito	0	0.6	7.2	USD
DynamoDB on-demand capacity	0	2.85	34.2	USD
DynamoDB provisioned capacity	205.2	29.83	563.16	USD
Amazon EC2	0	1.022	12.26	USD
Amazon EventBridge	0	0	0	USD
Standard Text-to-Speech	0	0	0	USD
SageMaker Training	0	2.18	26.16	USD
S3 Standard	0	0	0	USD
Data Transfer	0	0	0	USD
Amazon Transcribe	0	0	0	USD
AWS Lambda	0	0	0	USD

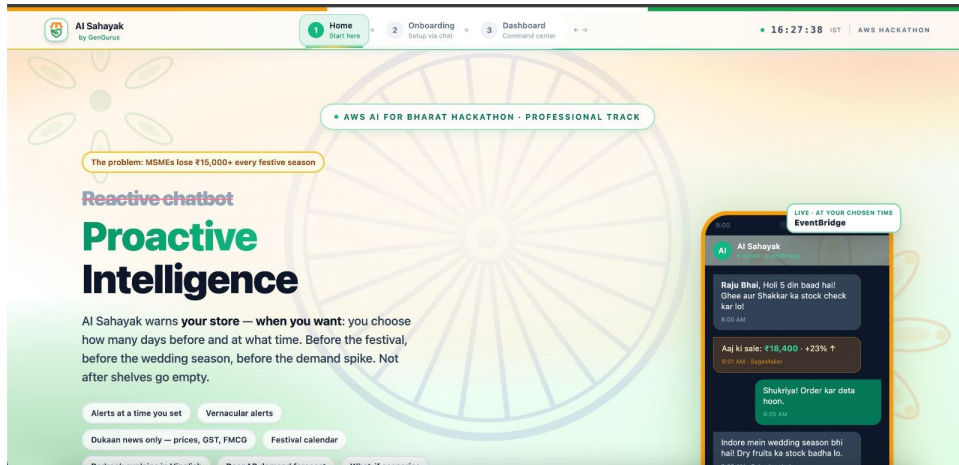
Snapshots of the prototype:

1



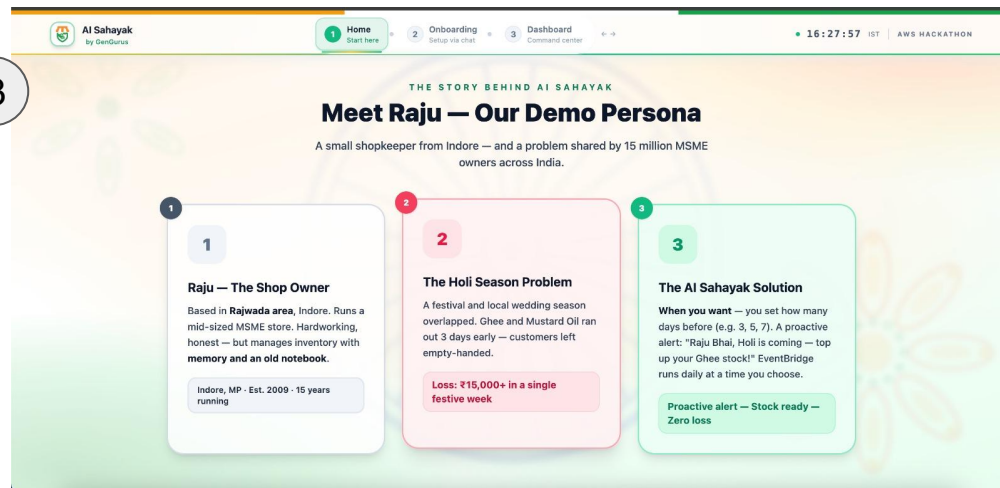
Logo meaning screen: Introduces AI Sahayak branding, color symbolism, and the core promise before entering the product.

2



Home value proposition: Establishes the shift from reactive reports to proactive festival-time intelligence with timed alerts.

3



Persona story (Raju): Explains the MSME pain point and how AI Sahayak prevents festive stockout losses.

4

HOW WE ARE DIFFERENT

Not Reactive — Proactive

Other tools only show you data. We alert you *proactively* — before the loss happens.

— THE OLD WAY
Reactive

- ✗ Check report → stock already empty
- ✗ Festival arrives → realise too late
- ✗ Customer leaves → only then reorder
- ✗ Pure guesswork — no confidence score

+ AI SAHAYAK
Proactive

- ✓ EventBridge: daily check at a time you set
- ✓ Proactive alert — you choose how many days before
- ✓ SageMaker DeepAR: demand forecast
- ✓ Bedrock: explains why, in plain language

Reactive vs proactive comparison: Clearly contrasts old workflow failures against AI Sahayak's prediction-first approach.

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POWERED BY AWS

The AWS tech stack powering this

Lambda — Festival check

EC2 — Compute

DynamoDB — Store data

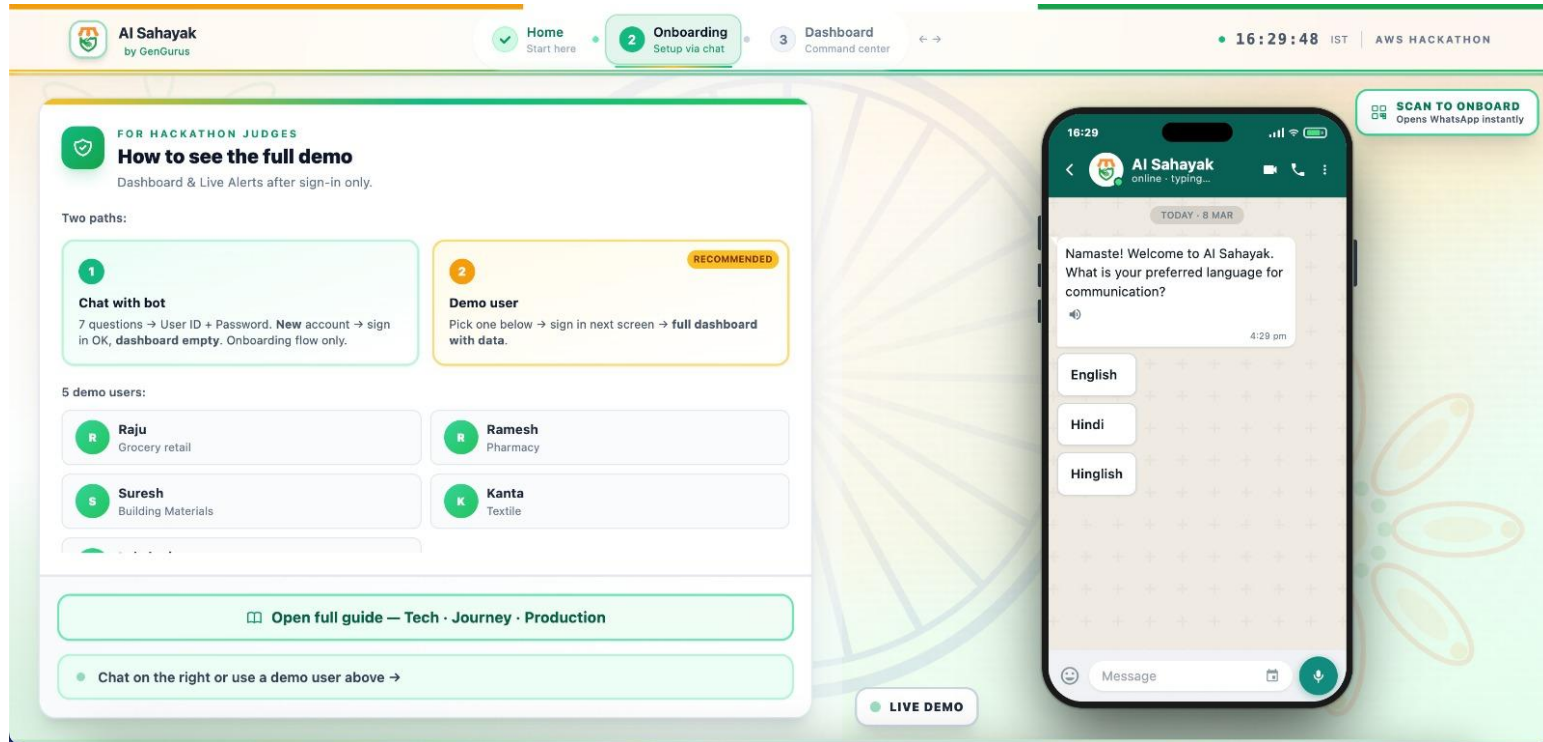
S3 — Festival calendar

Bedrock — Hinglish AI

SageMaker — Demand forecast

Ev

AWS stack highlight: Shows the core AWS services powering forecasting, alerts, data, and reasoning.



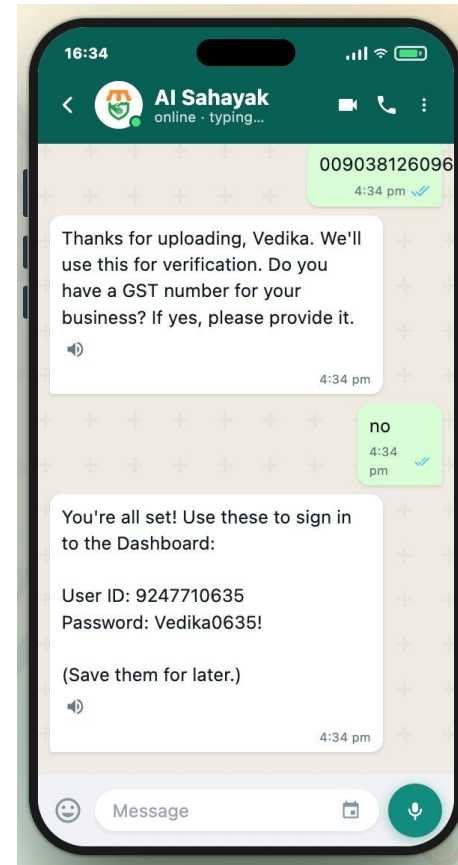
Onboarding entry screen: Lets users choose guided chat onboarding or instant demo-user access for judges.

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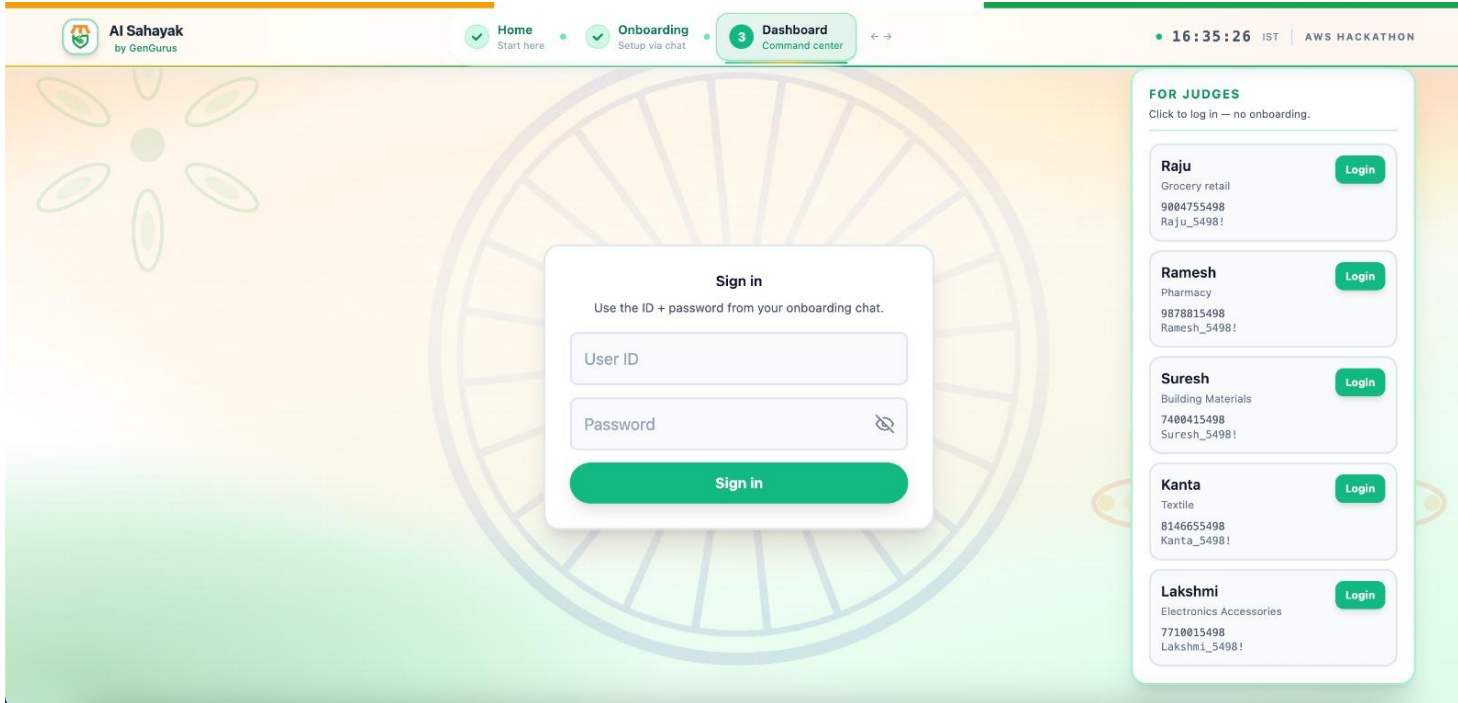


WhatsApp onboarding (early steps): Captures language and identity details conversationally through chat.

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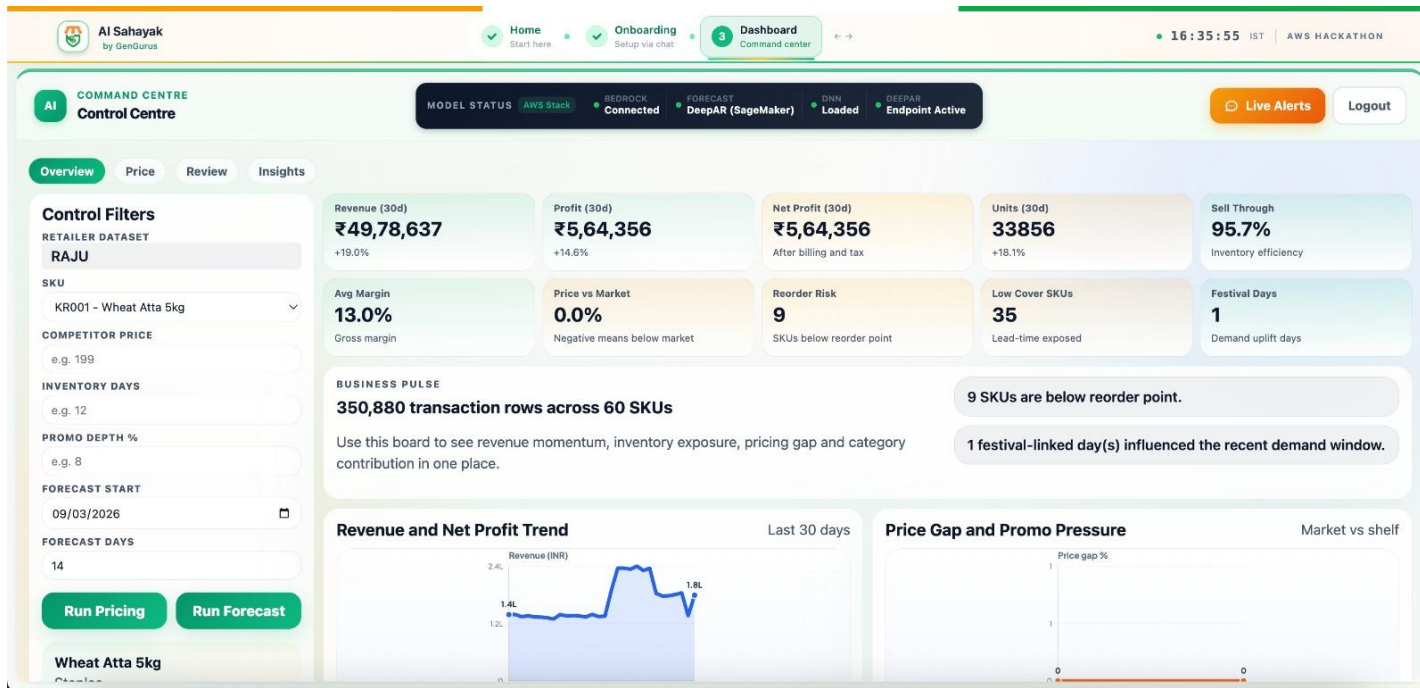


WhatsApp onboarding (completion): Confirms verification and provides dashboard login credentials to the user.



Dashboard sign-in screen: Supports manual login and one-click judge demo-user login shortcuts.

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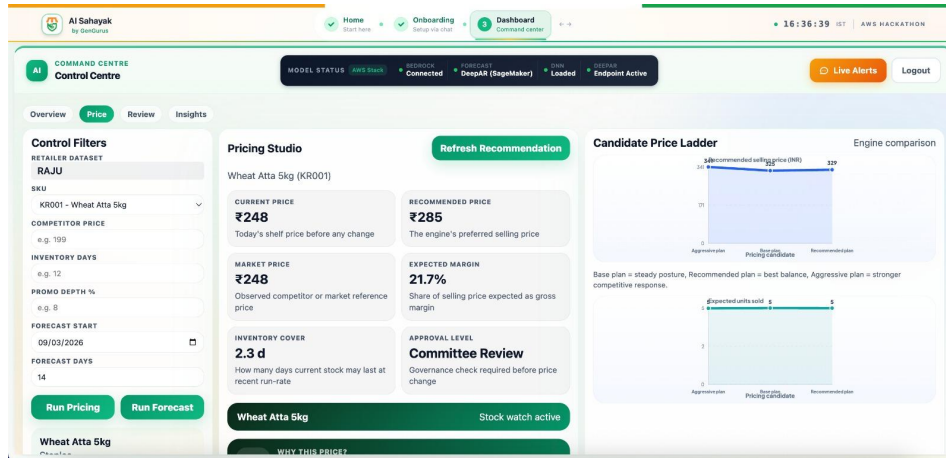
Control Centre overview: Presents KPI cards, filters, business pulse, and trend panels in one command view.

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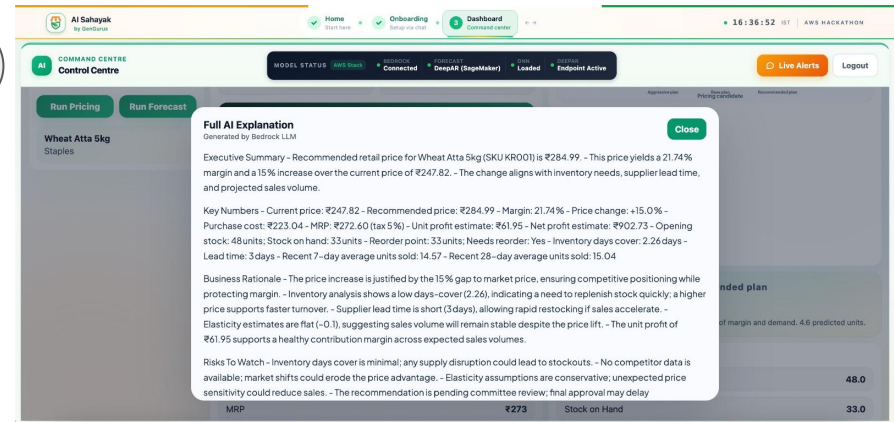
Model health strip: Displays real-time status of Bedrock, DeepAR/SageMaker, and endpoint readiness.

12



Pricing Studio view: Shows AI-recommended price, margin impact, and candidate ladder for decision support.

13



AI rationale modal (base case): Gives Bedrock-generated executive summary, key numbers, and risk notes behind pricing.

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AI Sahayak
by dardurax

Home Start here Onboarding Setup via chat **Dashboard Command center** 16:37:26 IST AWS HACKATHON

COMMAND CENTRE
Control Centre

MODEL STATUS AWS Stack Bedrock Connected Forecast DeepAR (SageMaker) Data Loaded Deploy Endpoint Active Live Alerts Logout

COMPETITOR PRICE
e.g. 199

INVENTORY DAYS
e.g. 12

PROMO DEPTH %
e.g. 8

FORECAST START
09/03/2026

FORECAST DAYS
14

Run Pricing **Run Forecast**

Wheat Atta 5kg
Staples

Recommended price ₹285

Item Wheat Atta 5kg

Margin 21.7%

What-If Scenario Evaluate

During the Holi rush, competitor drops price to 95, stock cover is only 4 days and promo depth should be 8%

Type natural language. The engine parses competitor moves, stock stress, promo depth and event context.

SAHAYAK'S ANSWER
Decision - Set the selling price of Wheat Atta 5kg (SKU KR001)
Explained by Bedrock LLM

[Open Full Explanation](#)

Parsed Inputs How the system understood your request

```
{
  "channel": "GT",
  "competitor_price": 95,
  "inventory_days": 4,
  "promo_depth_pct": 8,
  "sku_id": "KR001"
}
```

What-if simulation input: Accepts natural-language scenarios and shows parsed structured inputs plus decision output.

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AI Sahayak
by dardurax

Home Start here Onboarding Setup via chat **Dashboard Command center** 16:37:35 IST AWS HACKATHON

COMMAND CENTRE
Control Centre

MODEL STATUS AWS Stack Bedrock Connected Forecast DeepAR (SageMaker) Data Loaded Deploy Endpoint Active Live Alerts Logout

COMPETITOR PRICE
e.g. 199

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FORECAST DAYS
14

Run Pricing **Run Forecast**

Wheat Atta 5kg
Staples

Full AI Explanation
Generated by Bedrock LLM

Executive Summary - Holi rush triggers a competitor price cut to ₹95 for Wheat Atta 5kg. - Current inventory covers only 4 days; a promotional depth of 8% is recommended. - Updated recommendation: set selling price at ₹255.07, reducing margin to 12.56% but preserving volume.

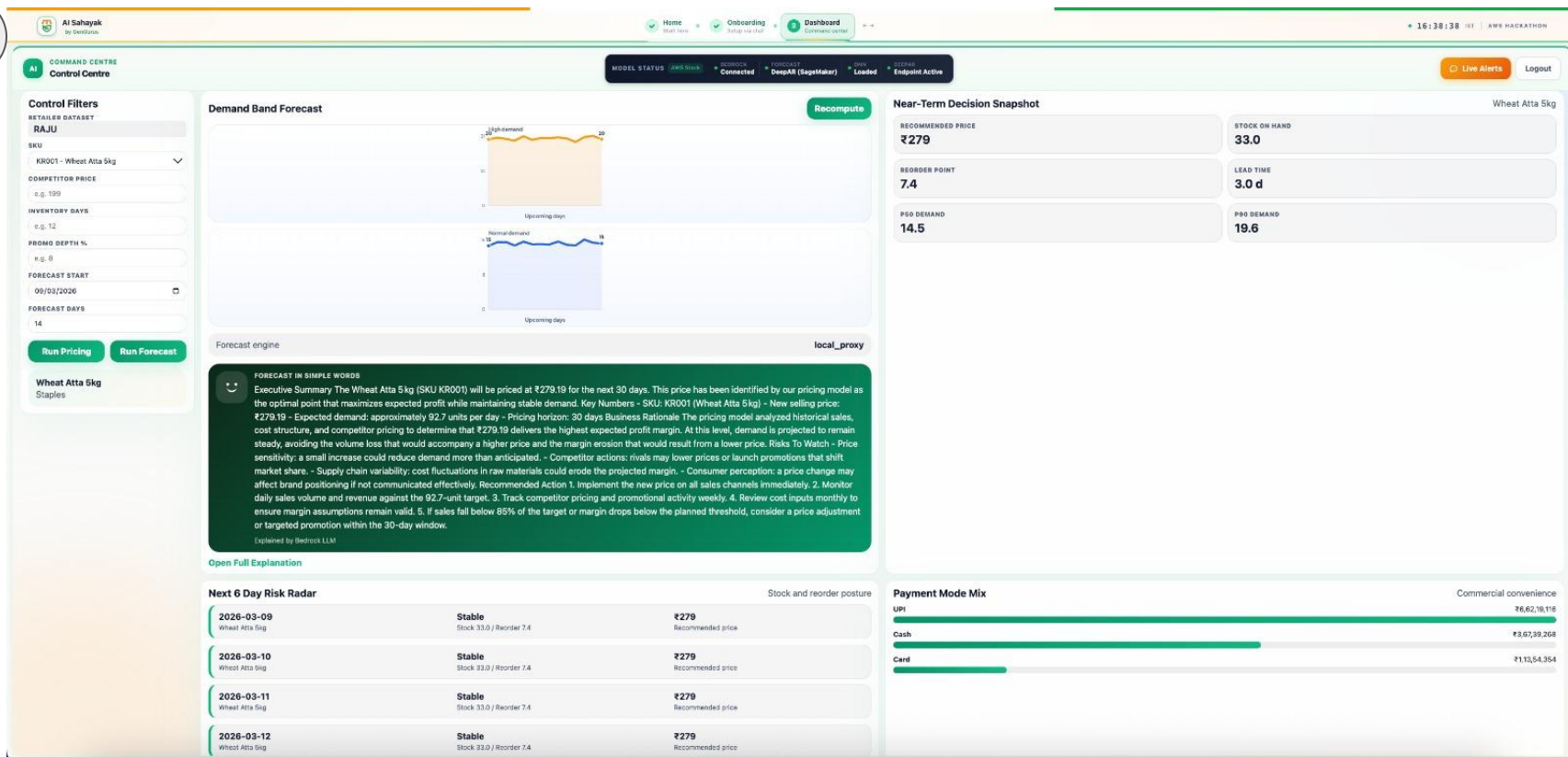
Key Numbers - SKU: KR001 - Wheat Atta 5kg (Staples) - Current price: ₹247.82 - Competitor price: ₹95.00 (gap 168.49%) - Recommended price: ₹255.07 (change +2.93%) - Margin: 12.56% (down from 21.74%) - Promo depth: 8% - Inventory days cover: 2.26 days (needs reorder) - Lead time: 3 days - Unit profit estimate: ₹32.03 (down from ₹61.95) - Net profit estimate: ₹466.75 (down from ₹902.73)

Business Rationale - Competitor's aggressive pricing forces a price adjustment to stay competitive. - Maintaining a modest 8% discount keeps the product attractive while protecting some margin. - The 4-day stock cover indicates urgency; a lower price will accelerate sales and reduce stock-out risk. - The price shift aligns with the "price_optimal" source, ensuring data-driven decision making.

Risks To Watch - Margin erosion could impact overall profitability if volume does not rise proportionally. - Rapid price change may confuse loyal customers; clear in-store signage is essential. - Inventory turnover may still be insufficient; monitor sales daily and be ready to reorder within 3 days. - Competitor may further lower prices; maintain flexibility to adjust within the 8% promo band.

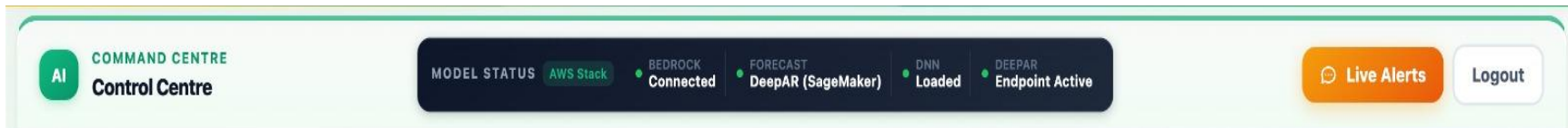
[Open Full Explanation](#)

AI rationale modal (what-if case): Explains updated price recommendation under competitor pressure and stock constraints.



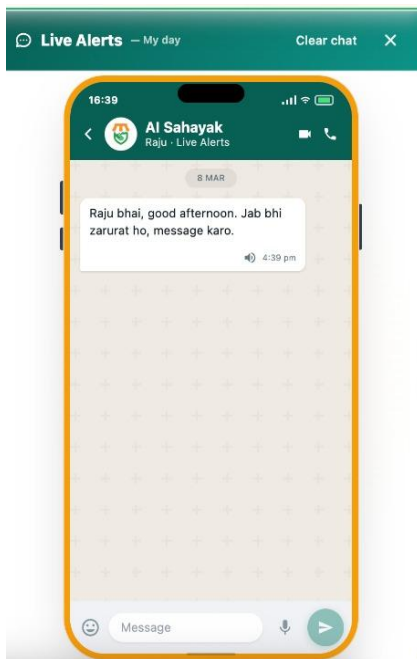
Forecast insights screen: Combines demand bands, decision snapshot, risk radar, and payment-mode mix for execution.

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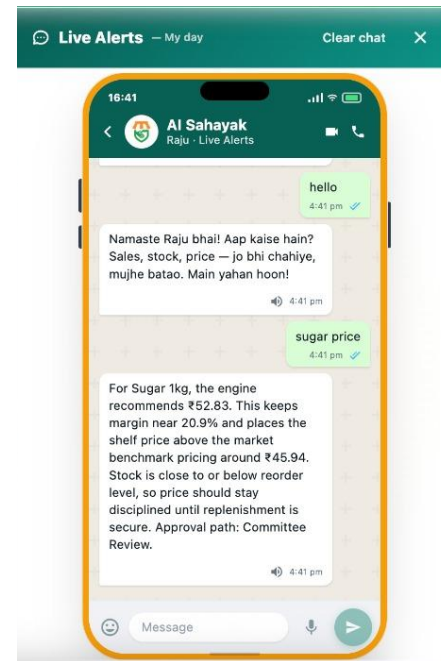
Command Centre header state: Highlights navigation, model status, and quick controls (Live Alerts/Logout).

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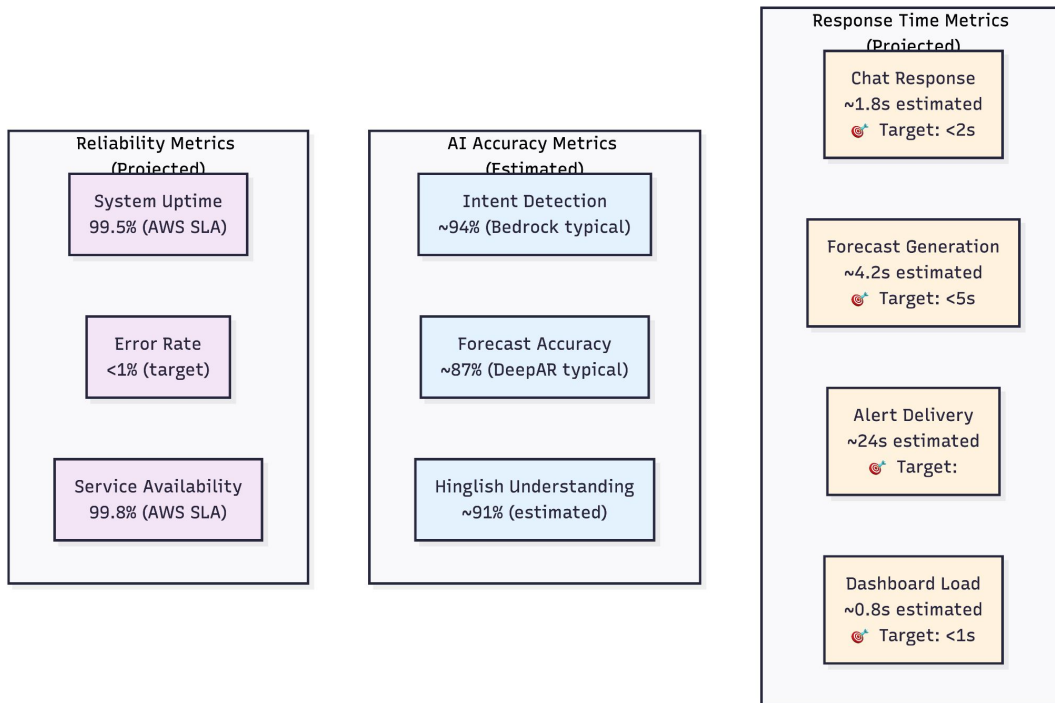
Live Alerts side panel: Demonstrates proactive WhatsApp alert delivery directly alongside the dashboard.

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Interactive alert/chat loop: Shows real-time user query handling with grounded product-specific pricing advice.

Prototype Performance report/Benchmarking:



Additional Details/Future Development :

Seamless WhatsApp Ecosystem

- **Zero-Friction UX:** Onboarding, daily reports, and alerts happen entirely within WhatsApp—no app switching required for the shopkeeper.
- **Unified Interface:** The Judge's Dashboard serves as the real-time "back-office" view of all WhatsApp-driven data.

Intelligent OCR (Powered by Amazon Textract)

- **Khatabook Integration:** Shopkeepers can simply snap a photo of their physical ledger or a screenshot of the Khatabook app.
- **Automated Data Flow:** Textract extracts sales, *udhaar*, and payment data → AI Sahayak processes → Dashboard & forecasts update automatically.
- **Smart Invoicing:** Snap receipts or purchase bills to auto-update inventory and line-item expenses.

The Road Ahead

- **Real-time Insights:** A photo sent on WhatsApp triggers a **Bedrock-powered** extraction, updating Control Center graphs in near real-time.
- **Voice-First Interface:** "Batao aaj ka summary"—Multilingual TTS (Text-to-Speech) for hands-free operations.
- **Financial Sync:** Direct API/CSV integration with Khatabook for automated credit reminders and outstanding payment tracking.
- **Scalability:** Multi-store management for owners with various retail locations through a single chat interface.

Prototype Assets:

- GitHub Public Repository Link : [AI-Sahayak GitHub](#)
- Demo Video Link : [AI-Sahayak Demo Video](#)

Innovation partner **I12S**
HACKATHON

Media partner **YOURSTORY**

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Thank You

